

PE01: PE of 07/11/2020 — Solutions

Master in Informatics and Computing Engineering
Programming Fundamentals
Instance: 2020/2021

An example of solutions for the 5 questions in this Practical on computer evaluation.

1. If odd or even

Write a Python script that receives three integers n1, n2 and n3 given by user input.

- If n1 is odd then the output is the product of n2 and n3.
- If n1 is even, then the output is the subtraction between n2 and n3.

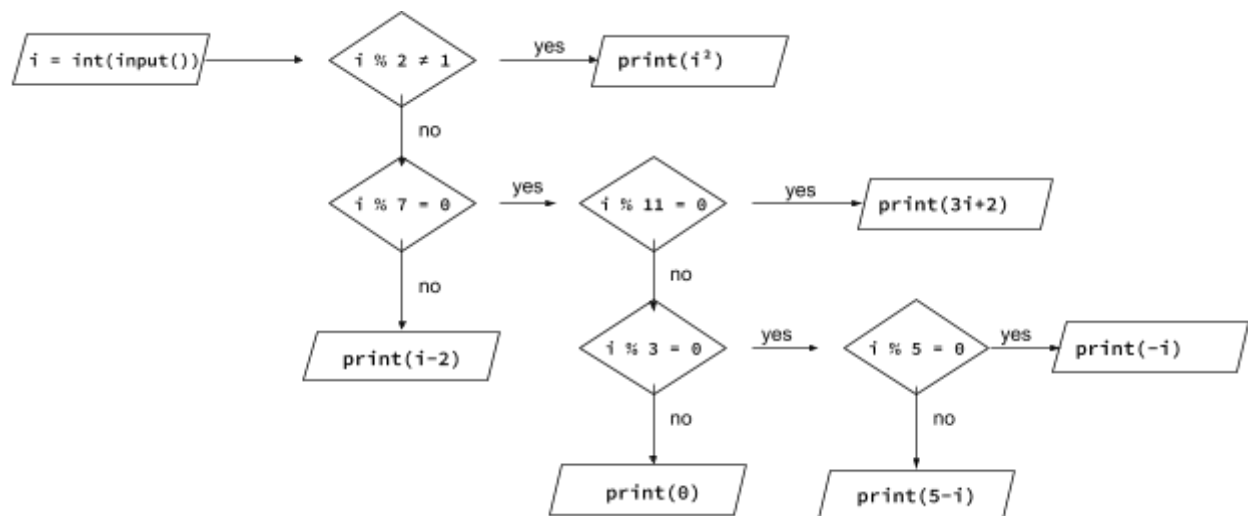
Solution:

```
n1 = int(input("n1? "))
n2 = int(input("n2? "))
n3 = int(input("n3? "))

if n1 % 2 != 0:
    print(n2*n3)
else:
    print(n2-n3)
```

2. Flowchart

Implement the following flowchart as a Python script:



Solution:

```
i = int(input("i? "))
if i % 2 != 1:
    print(i**2)
elif i % 7 == 0:
    if i % 11 == 0:
        print(3*i+2)
    elif i % 3 == 0:
        if i % 5 == 0:
            print(-i)
        else:
            print(5-i)
    else:
        print(0)
else:
    print(i-2)
```

3. Volumes

Write a Python script that asks the user for a shape and all necessary parameters, then prints the volume of the shape, rounded to 1 decimal.

For example, if the user requests a cylinder, you must then ask for its radius and height.

The possible requested shapes, the shape's parameters, and corresponding output are the following:

Shape (str)	Parameters (float)	Output
"sphere"	radius (r)	$\frac{4}{3}\pi r^3$
"cylinder"	radius (r), height (h)	$\pi r^2 h$
"cone"	radius (r), height (h)	$\frac{1}{3}\pi r^2 h$

Solution:

```
import math

shape = input("shape? ")
r = float(input("r? "))

if shape == 'sphere':
    result = (4/3)*math.pi*(r**3)
if shape == 'cylinder':
    h = float(input("h? "))
    result = math.pi*(r**2)*h
if shape == 'cone':
    h = float(input("h? "))
    result = (1/3)*math.pi*(r**2)*h

print(round(result, 1))
```

4. Multiplicative persistence

The *multiplicative persistence of a number* is the number of times the digits of a positive integer can be multiplied until the number has only one digit.

For example, the persistence of the number 39 is 3 because

$$39 \rightarrow 3 \cdot 9 = 27 \rightarrow 2 \cdot 7 = 14 \rightarrow 1 \cdot 4 = 4.$$

Write a Python script that, given a number `num` by user input, prints out its multiplicative persistence.

Solution:

```
num = int(input("num? "))

count = 0
while num > 9:
    aux = 1
    while num != 0:
        aux = (num % 10) * aux
        num = num // 10
    num = aux
    count += 1

print(count)
```

5. Only while and if

The upcoming Python XP is a new cleaned-up version and many flow control statements have been removed.

Modify the following code so that only while and if are used as flow control (i.e., you cannot use for, elif, else, continue or break).

```
a = int(input())
b = int(input())

for i in range(a, b, 3):
    n = i
    if i**2 > (b/2)**2:
        break
    if i % 3 == 0 and i % 5 == 0:
        continue
    elif i % 2 == 0:
        for _ in range(3):
            n //= 2
    else:
        n += 1
    print(n)
```

Solution:

```
a = int(input("a? "))
b = int(input("b? "))

i = a
while i < b and not (i**2 > (b/2)**2):
    n = i
    if not (i % 3 == 0 and i % 5 == 0):
        if i % 2 == 0:
            n //= 2
            n //= 2
            n //= 2
        if not (i % 2 == 0):
            n += 1
    print(n)
    i += 3
```

The end.
